INVERTIS UNIVERSITY BAREILLY
Established by Govt. of U.P. u/s 2f of UGC Act, 1956 vide U.P. Act 22 of 2010.

BCS-401	Computer Organization & Architecture	L T P 3 0 0	3 credits

Course Objectives:

CO1	Conceptualize the basics of organizational and architectural issues of a digital computer.
CO2	Understand concepts of register transfer logic and arithmetic operations.
CO3	Understand the various computer architectures and control units
CO4	How I/O devices are accessed and its principles.
CO5	The current state of art in memory system design

MODULE-I

Introduction:, Digital computer Block diagram, functional units and their interconnections, buses, types of buses and bus arbitration.

Number representation : Fixed point Integer representation , Fixed point arithmetic operations in 2's complement form: Addition, Subtraction, Booths multiplication algorithm, array multiplier, and Division techniques. Floating point number representation, IEEE standard for floating point representation, Floating point arithmetic operation .

MODULE-II

Central Processing unit: Register, bus and Memory transfer, Register Transfer language, Arithmetic, logic and shift micro operations, arithmetic and logic unit, Processor organization: Single Accumulator, general register and stack organization, Addressing modes, Instruction types, Instruction formats, instruction cycle.

Control Unit :Hardwired and microprogrammed control, concept of horizontal and vertical microprogramming.

MODULE-III

Memory: Basic concept and hierarchy, semiconductor RAM memories, ROM memories, Cache memory, address mapping techniques and replacement, Auxiliary memories, Virtual memory.

Input / Output: Peripheral devices, I/O interface, I/O ports, Interrupts, types of interrupts, Modes of Data Transfer, Programmed I/O, interrupt initiated I/O and Direct Memory Access.
Pipelining: Basic concepts of pipelining, throughput and speedup.

Text Books:

1. William Stalling- Computer Organization and Architecture,PHI
2. Morris Mano, Computer System Architecture, PHI

Reference Books:

1. "Computer Organization and Design: The Hardware/Software Interface" , 5th Edition by David A. Patterson and John L. Hennessy, Elsevier.
2. John P. Hays – Computer Organization , Mc-Graw Hill
3. Vravice,Hamacher & Zaky, "Computer Organization", TMH

Course Outcomes:

CO1	Draw the functional block diagram of a single bus architecture of a computer and describe the function of the instruction execution cycle, RTL interpretation of instructions, addressing modes, instruction set.
CO2	Define different number systems, binary addition and subtraction, 2's complement representation and operations with this representation.
CO3	Understand the architecture and functionality of central processing unit.
CO4	Exemplify in a better way the I/O and memory organization.
CO5	Given a CPU organization and instruction, design a memory module and analyze its operation by interfacing with the CPU.